

The burning question: How quickly do forbs respond to fire suppression?

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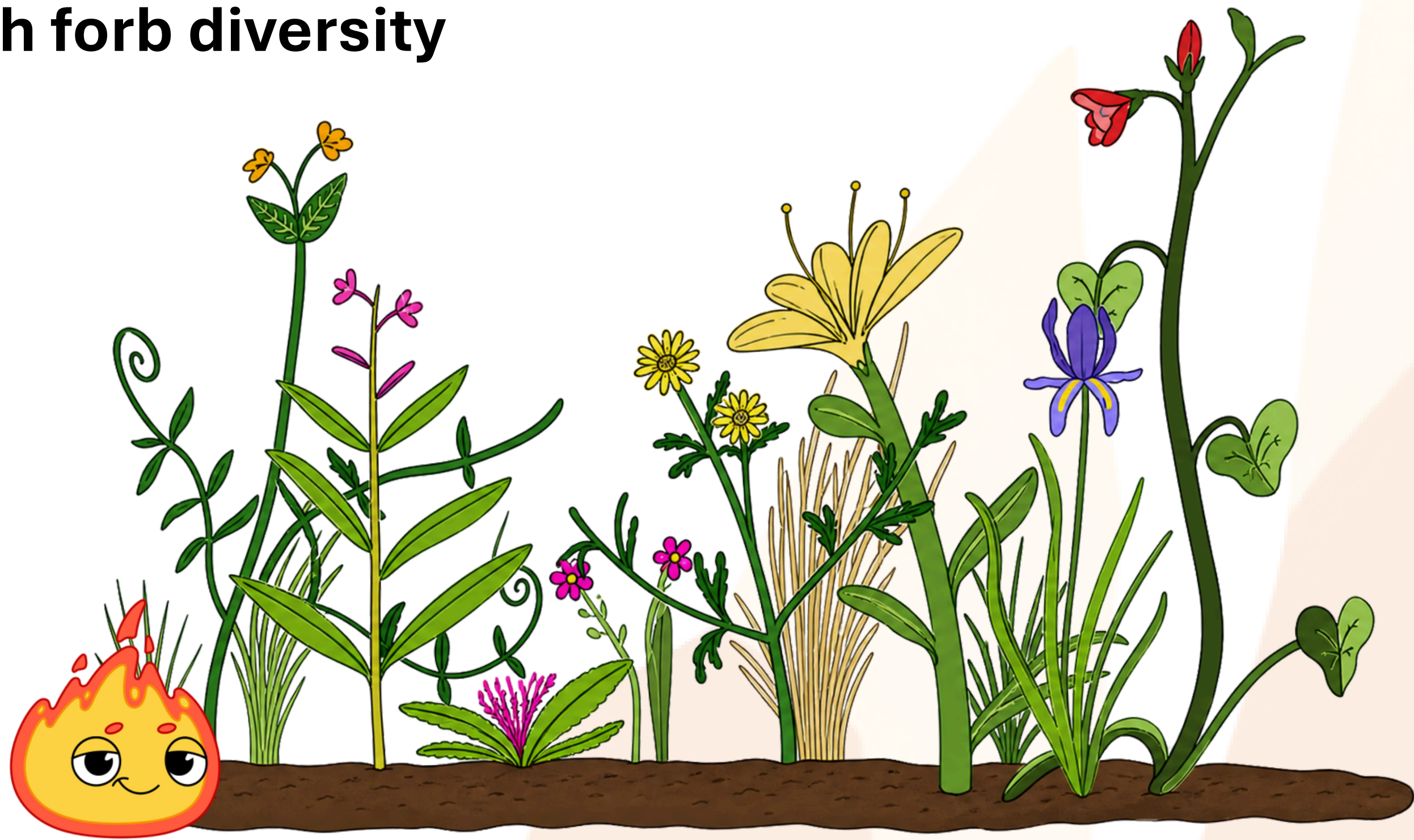
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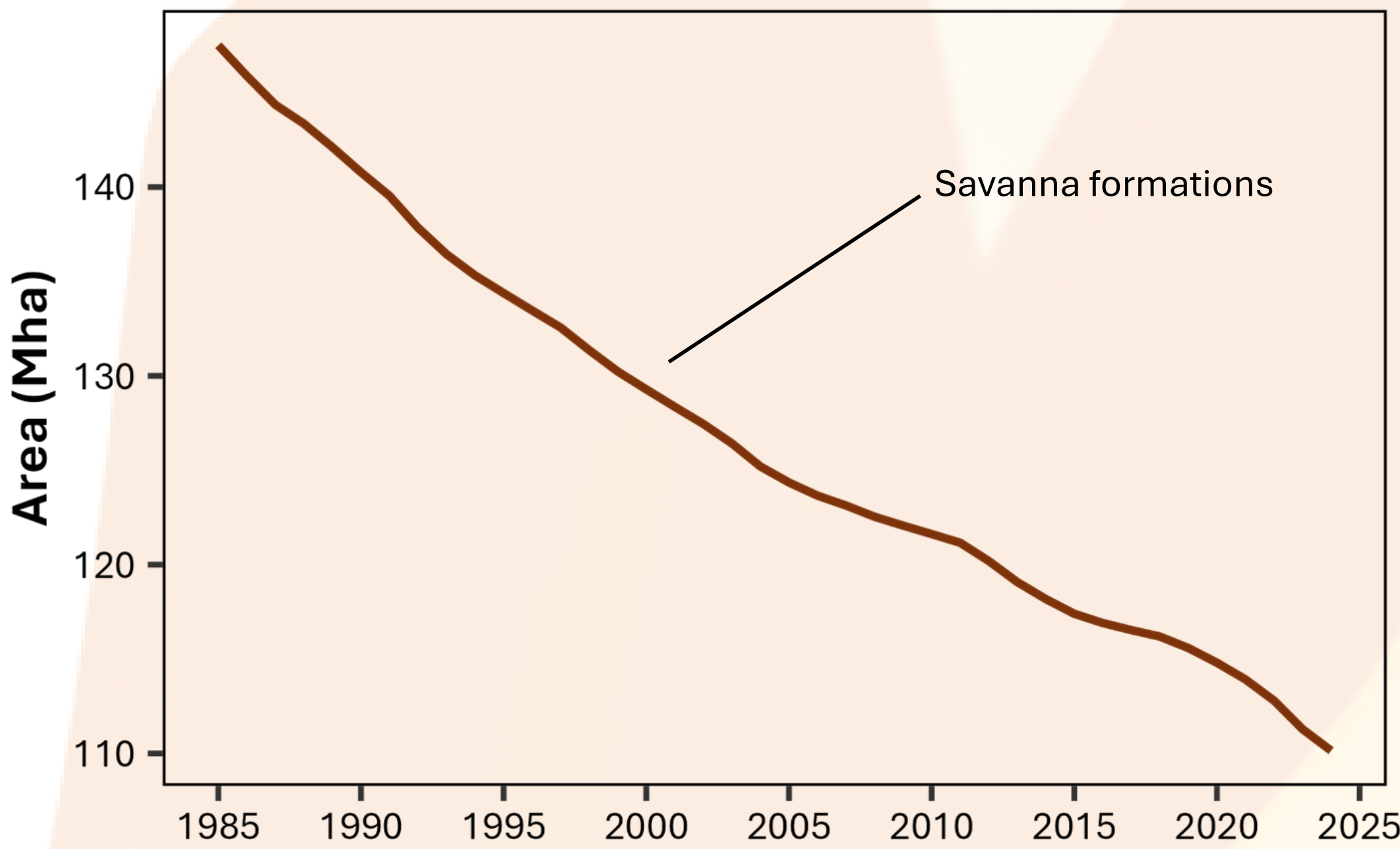
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Context and research question

Frequent fire maintains open Cerrado vegetation and **high forb diversity**

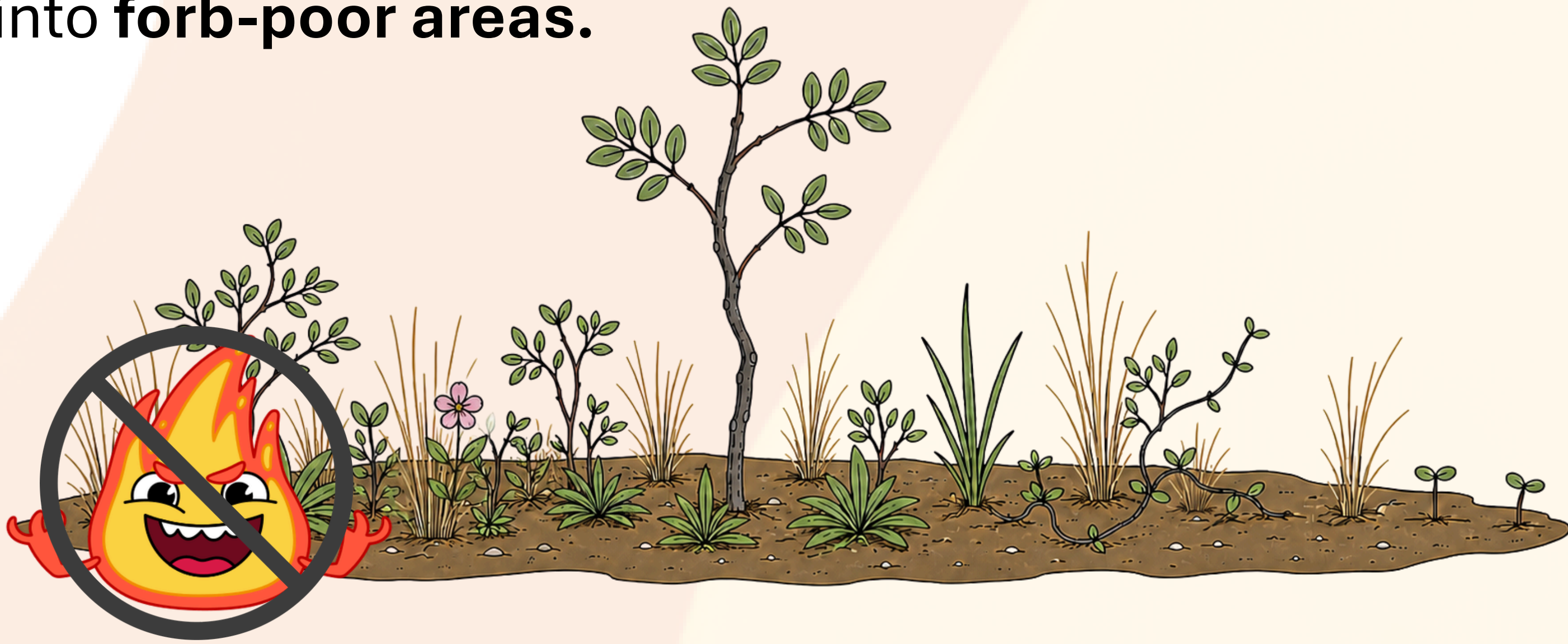


However, in Brazil, a **long-standing zero-fire policy** promoted **fire suppression for decades**, decreasing savanna formations and affecting **forb diversity**



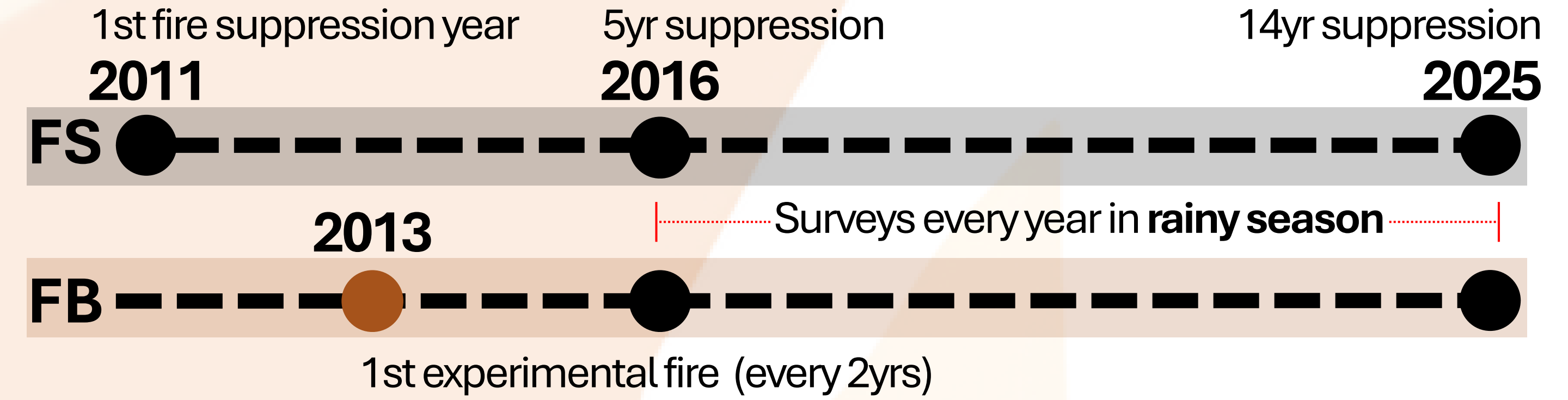
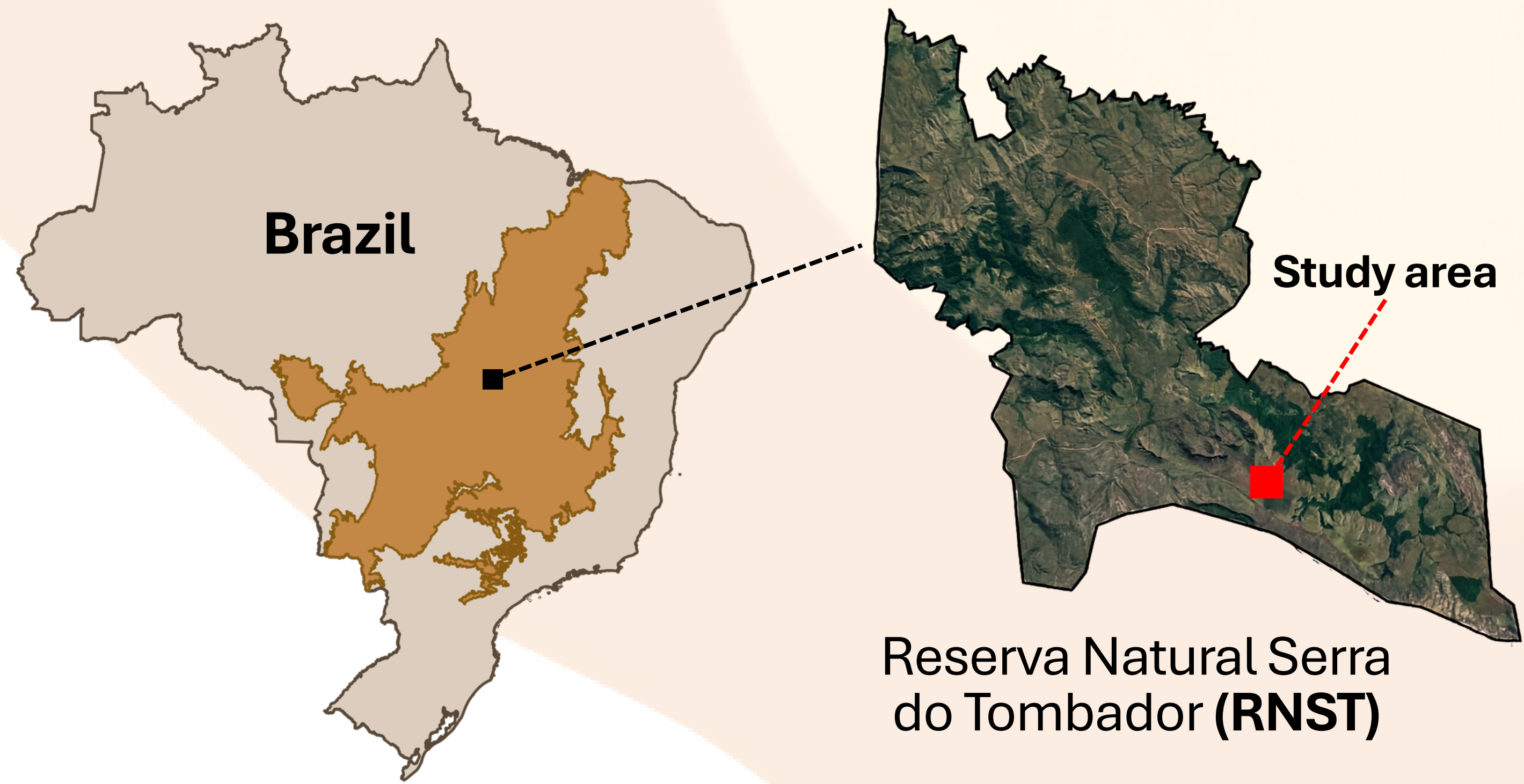
Trends in savanna formations in the Cerrado over the past 40 years (Mapbiomas)

We already know that this vegetation shift **leads to woody encroachment**, turning diverse communities into **forb-poor areas**.

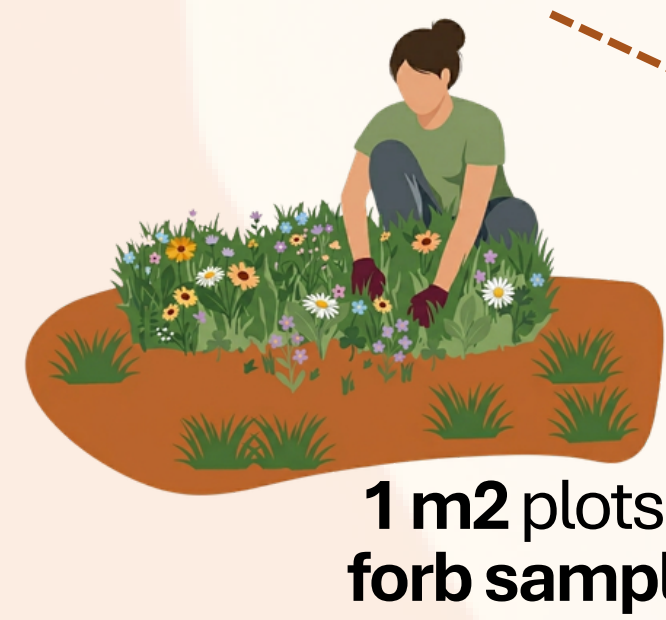


But when do changes in forb species actually become detectable?

Study area and methodology



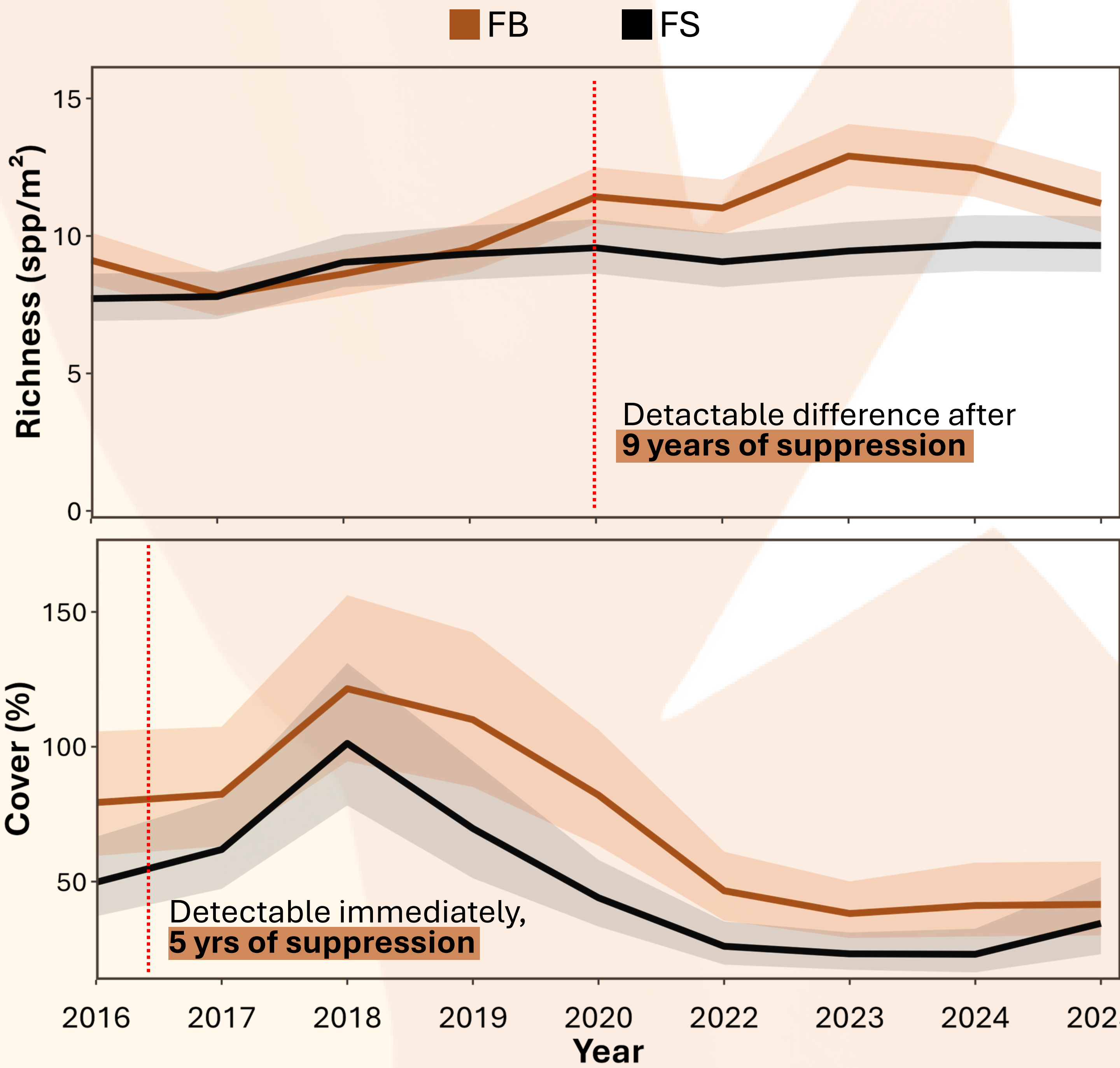
Scan for code & analysis



1 m² plots for forb sampling

3 plots (30 x 30 m) per treatment
10 subplots in each plot

Results and message



Different temporal responses:

Cover responded immediately, reaching values up to **87% higher** under frequent fire. Richness responded later, with species loss becoming apparent **after 9 years of suppression**.

Don't wait for species loss: early drops in forb cover can show biodiversity change years before richness declines



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